

Ornob Barua

(408) 375-7225 | ornobmb@gmail.com | <https://www.linkedin.com/in/ornob-barua/>

Portfolio link: <https://sites.google.com/view/ornob-barua-portfolio/home>

EDUCATION

B.S. Mechanical Engineering, University of California, Santa Barbara

Expected: June 2027

GPA: 3.84 Dean's Honors

Related Coursework: MEMS, Control System Design, Thermodynamics, Finite Element Analysis, Fluid Mechanics, Statics, Dynamical Systems, Machine Learning, Mechatronics, Vibrations, Heat Transfer, Design Optimization

EXPERIENCE

UCSB Nanofabrication Facility

March 2025 – Present

Processing Technician Intern

Goleta, California

- Operated scanning electron microscope (SEM) for imaging and analysis of different calibrations
- Performed FICP Si and SiO₂ calibrations using profilometry and ellipsometry to calculate selectivity and etch rate
- Executed photolithography using an ASML DUV stepper, followed by SEM analysis to verify pattern fidelity
- Performed ALD calibrations for Al₂O₃ and SiO₂ thin films, thickness verification with ellipsometry
- Completed calibrations for ICP-PECVD machine for SiO₂ and SiN thin films using metrology for analysis
- Improved maskless aligner (MLA) calibration process by redesigning calibration pattern with KLayout and Python, performing microscope and SEM-based analysis, and creating a traveler to guide new interns

Arista Networks

June 2025 – September 2025

Mechanical Engineer Intern

San Jose, California

- Designed and fabricated custom thermal test fixture for busbar power and return feeders
- Developed signal integrity test fixture for Time Domain Reflectometry (TDR) measurements of line card and fabric module systems, enabling reliable debugging
- Utilized Creo Parametric and sheet metal design practices to model components effectively for test fixture
- Applied DFM principles and tolerance analysis to fixture designs, reducing rework and ease of assembly
- Created detailed part and assembly engineering drawings and issued drawings to vendors for quotation
- Tested Arista 7800 installation guide through full chassis assembly to improve clarity and usability

Cadence Group, UCSB

October 2024 – June 2025

Undergraduate Research Assistant

Goleta, California

- Developed a synthetic shoe aging machine, collaborating with small team to meet project specifications
- Engineered a sensor system using an Arduino Nano and reed switch to quantify shoe aging in a reliable manner
- Designed a custom motor bracket and wood chassis with SolidWorks to ensure motor stability during operation
- Implemented electrical housing for the motor's power supply and sensor system to ensure safe operation

Microsoft

June 2024 - September 2024

Garage Makerspace Intern

Mountain View, California

- Provided technical support at the Microsoft Garage, assisting with prototyping and maintaining all equipment
- Co-facilitated and developed hands-on workshops integrating Arduino, 3D printing, and robotics, including projects such as Automatic Water Feeder, Line following robot, Silicon hand, Self-balancing robot, and RFID safe
- Produced concise and accurate prompts for Copilot to create and optimize Arduino code for all workshops

Projects

SiO₂/Glass/PDMS Microchannels

March 2026 – June 2026

- Developed and followed process followers utilizing thermal oxidation, wet etching, contact photolithography, ellipsometer, and profilometer to fabricate 3 different types of microchannels
- Analyzed data from microscopy, ellipsometry, and profilometry to fabricate chips to target dimensions
- Conducted current monitoring with electrode set up to test microchannels for microfluidic flow by applying 100V to 500V with channel filled with electrolyte solution and compare to analytical results

Automated Toilet Paper Replacement System

March 2026 – June 2026

- Engineered an electromechanical system integrating Arduino-based motor control, IR sensing, and custom laser-cut and 3D-printed components
- Optimized mechanism and control sequence to achieve automated replacement in 4.73 seconds

SKILLS

Technical: Creo Parametric, SolidWorks, Fusion, GD&T, Cleanroom Manufacturing and Metrology, SEM, 3D printing, Arduino, MATLAB, COMSOL, Python, KLayout, Laser Cutting, Soldering, LaTeX, MS Office Suite

Professional: Teamwork, Time management, Adaptability, Problem Solving, Presenting & Articulation